CSV content analysis

- number of URLs in the content
- number of dates in the content
- number of numeric values in the content
- the 100 most frequent words in the content
- a plot of the 10,000 most frequent words

```
from pathlib import Path
import re
from collections import Counter
import time

import duckdb
import pandas as pd
import matplotlib.pyplot as plt
```

1. Set the CSV path

Change the path below so it points to your CSV file.

```
CSV_PATH = Path(r"C:\Users\jespe\GDS eksamen\995,000_rows.csv")
CONTENT_COLUMN = "content"
ENCODING = "utf-8"
CHUNK_SIZE = 500000  # Process 500K rows at a time
```

2. Validate CSV header

This checks that the file is readable and that the target column exists, without loading the full dataset.

```
header_df = pd.read_csv(CSV_PATH, encoding=ENCODING, nrows=0)

if CONTENT_COLUMN not in header_df.columns:
    raise ValueError(
        f"Column '{CONTENT_COLUMN}' was not found. Available columns:
{list(header_df.columns)}"
    )

print(f"CSV is readable: {CSV_PATH}")
print(f"Detected columns: {list(header_df.columns)}")
print(f"Using chunk size: {CHUNK_SIZE:,} rows")
```

```
CSV is readable: C:\Users\jespe\GDS eksamen\995,000_rows.csv
Detected columns: ['Unnamed: 0', 'id', 'domain', 'type', 'url',
'content', 'scraped_at', 'inserted_at', 'updated_at', 'title',
```

```
'authors', 'keywords', 'meta_keywords', 'meta_description', 'tags',
'summary', 'source']
Using chunk size: 500,000 rows
```

3. Regex patterns

These patterns are practical defaults:

- **URL pattern**: matches `http://`, `https://`, and `www.`
- **Date pattern**: matches common formats such as:
  - `2024-03-11`
  - `11-03-2024`
  - `11/03/2024`
  - `March 11, 2024`
- **Numeric pattern**: matches integers, decimals, and signed values

```
URL_PATTERN = re.compile(
    r"\b(?:https?://|www\.)[^\s<>'\"()]+",
    flags=re.IGNORECASE
)

DATE_PATTERN = re.compile(
    r"\b(?:"
    r"\d{1,2}[/-]\d{1,2}[/-]\d{2,4}"
    r"|"
    r"\d{4}[/-]\d{1,2}[/-]\d{1,2}"
    r"|"
    r"(?:jan|feb|mar|apr|may|jun|jul|aug|sep|sept|oct|nov|dec)"
    r"[a-z]*\s+\d{1,2},?\s+\d{2,4}"
    r")\b",
    flags=re.IGNORECASE
)

NUMERIC_PATTERN = re.compile(
    r"(?<!\w)[+-]?(?:\d+(?:[.,]\d+)?)(?!\w)"
)

WORD_PATTERN = re.compile(r"\b[a-zA-ZÀ-ÿ']+\b")
```

4. Stream through CSV and aggregate counts

This processes the file chunk-by-chunk to avoid loading the full dataset into memory.

```
start_time = time.time()
print(f"Starting DuckDB processing at {time.strftime('%H:%M:%S')}\n")

# Create DuckDB connection with memory limit
con = duckdb.connect()
```

```
con.execute("SET memory_limit='8GB'")
con.execute("SET threads=8")

print("Step 1: Counting total rows...")
scan_start = time.time()

# Query directly without creating table to save memory
total_rows = con.execute(f"""
    SELECT COUNT(*)
    FROM read_csv_auto('{str(CSV_PATH)}',
                        sample_size=50000,
                        parallel=true,
                        strict_mode=false,
                        ignore_errors=true)
    WHERE content IS NOT NULL
""").fetchone()[0]

print(f"  ✓ Total rows: {total_rows:,} (scanned in {time.time() -
scan_start:.1f}s)\n")

# Step 2: Count URLs, dates, and numeric values - stream without
loading into table
print("Step 2: Counting URLs, dates, and numeric values...")
regex_start = time.time()

url_pattern = r'https?://[^\s<>''\"()]+|www\.[^\s<>''\"()]+'
date_pattern = r'\d{1,2}[/-]\d{1,2}[/-]\d{2,4}|\d{4}[/-]\d{1,2}[/-]\
d{1,2}'
numeric_pattern = r'\b\d+\.?\d*\b'

# Stream and count in one pass
result = con.execute(f"""
    SELECT
        SUM(LENGTH(regexp_extract_all(content, '{url_pattern}', 0)))
as total_urls,
        SUM(LENGTH(regexp_extract_all(content, '{date_pattern}', 0)))
as total_dates,
        SUM(LENGTH(regexp_extract_all(content, '{numeric_pattern}',
0))) as total_numeric
    FROM read_csv_auto('{str(CSV_PATH)}',
                        sample_size=50000,
                        parallel=true,
                        strict_mode=false,
                        ignore_errors=true)
    WHERE content IS NOT NULL
""").fetchone()

total_urls = result[0] if result[0] else 0
total_dates = result[1] if result[1] else 0
total_numeric = result[2] if result[2] else 0
```

```
print(f"  ✓ Done in {time.time() - regex_start:.1f}s")
print(f"    - URLs: {total_urls:,}")
print(f"    - Dates: {total_dates:,}")
print(f"    - Numeric: {total_numeric:,}\n")

# Step 3: Extract and count words in chunks
print("Step 3: Extracting and counting words...")
print("  (Processing in chunks to manage memory...)")
word_start = time.time()

word_counts = Counter()
chunk_size = 500000

for offset in range(0, total_rows, chunk_size):
    chunk_start_time = time.time()

    # Stream words from CSV chunk
    words_df = con.execute(f"""
        SELECT regexp_extract_all(LOWER(content), '[a-zA-ZÀ-ÿ'']+', 0)
as words
        FROM read_csv_auto('{str(CSV_PATH)}',
                           sample_size=50000,
                           parallel=true,
                           strict_mode=false,
                           ignore_errors=true)
        WHERE content IS NOT NULL
        LIMIT {chunk_size} OFFSET {offset}
    """).fetchdf()

    # Count words - handle arrays properly
    for word_list in words_df['words']:
        if word_list is not None and len(word_list) > 0:
            word_counts.update(word_list)

    processed = min(offset + chunk_size, total_rows)
    elapsed = time.time() - word_start
    remaining = (elapsed / processed) * (total_rows - processed) / 60
if processed > 0 else 0

    print(f"    - Processed {processed:,}/{total_rows:,} rows
({processed/total_rows*100:.1f}%) "
          f"in {elapsed:.1f}s - Est. remaining: {remaining:.1f} min")

    del words_df

print(f"  ✓ Word counting complete in {time.time() - word_start:.1f}s\
n")

# Step 4: Create preview
```

```
print("Step 4: Creating preview...")
preview_df = con.execute(f"""
    SELECT
        content,
        LENGTH(regexp_extract_all(content, '{url_pattern}', 0)) as
url_count,
        LENGTH(regexp_extract_all(content, '{date_pattern}', 0)) as
date_count,
        LENGTH(regexp_extract_all(content, '{numeric_pattern}', 0)) as
numeric_count
    FROM read_csv_auto('{str(CSV_PATH)}',
                       sample_size=50000,
                       parallel=true,
                       strict_mode=false,
                       ignore_errors=true)
    WHERE content IS NOT NULL
    LIMIT 5
""").fetchdf()

total_time = time.time() - start_time
print(f"\n{'='*60}")
print(f"PROCESSING COMPLETE!")
print(f"{'='*60}")
print(f"Total rows processed: {total_rows:,}")
print(f"Total time: {total_time/60:.1f} minutes ({total_time/3600:.2f}
hours)")
print(f"Average speed: {total_rows/total_time:.0f} rows/second")
print(f"{'='*60}\n")

# Close connection
con.close()

preview_df
```

```
Starting DuckDB processing at 19:04:31

Step 1: Counting total rows...

<>:29: SyntaxWarning: invalid escape sequence '\.'
<>:29: SyntaxWarning: invalid escape sequence '\.'
C:\Users\jespe\AppData\Local\Temp\ipykernel_9108\1108022216.py:29:
SyntaxWarning: invalid escape sequence '\.'
  url_pattern = r'https?://[^\s<>''\"()]+|www\.[^\s<>''\"()]+'

  ✓ Total rows: 995,058 (scanned in 3.8s)

Step 2: Counting URLs, dates, and numeric values...
  ✓ Done in 7.1s
    - URLs: 269,539
    - Dates: 83,929
```

```
    - Numeric: 9,246,055

Step 3: Extracting and counting words...
  (Processing in chunks to manage memory...)
    - Processed 500,000/995,058 rows (50.2%) in 179.3s - Est.
remaining: 3.0 min
    - Processed 995,058/995,058 rows (100.0%) in 538.9s - Est.
remaining: 0.0 min
  ✓ Word counting complete in 602.3s

Step 4: Creating preview...

============================================================
PROCESSING COMPLETE!
============================================================
Total rows processed: 995,058
Total time: 10.2 minutes (0.17 hours)
Average speed: 1618 rows/second
============================================================


                                                content  url_count
date_count  \
0  Plus one article on Google Plus\n\n(Thanks to ...          0
0
1  The Cost Of The Best Senate Banking Committee ...          0
0
2  Man Awoken From 27-Year Coma Commits Suicide A...          0
0
3  WHEN Julia Geist was asked to draw a picture o...          0
0
4  — 100 Compiled Studies on Vaccine Dangers (Act...          0
0

   numeric_count
0              2
1              5
2              2
3              7
4            995
```

5. Show total counts for the whole dataset

```
print(f"Total URLs found: {total_urls:,}")
print(f"Total dates found: {total_dates:,}")
print(f"Total numeric values found: {total_numeric:,}")
```

```
Total URLs found: 269,539
Total dates found: 83,929
Total numeric values found: 9,246,055
```

6. Find the 100 most frequent words in the content

Word frequencies were aggregated during chunked processing above.

```
top_100_words = word_counts.most_common(100)
top_100_df = pd.DataFrame(top_100_words, columns=["word", "count"])

top_100_df
```

```
          word     count
0          the  26197217
1           to  12665105
2           of  12497559
3          and  11515006
4            a  10016572
5           in   8722310
6         that   5601978
7           is   5073471
8          for   4386815
9           on   3522241
10          it   3357914
11           s   3324067
12        with   2834158
13          as   2804255
14         was   2446471
15         are   2444393
16          by   2344952
17           i   2291734
18        this   2281746
19          at   2198249
20          be   2176130
21         not   2114237
22          he   2109271
23         you   2100645
24        have   2099335
25        from   2043114
26          an   1723904
27         has   1709606
28        they   1676544
29         but   1647773
30         his   1632199
31          we   1602266
32          or   1534999
33        said   1415584
34         who   1347082
35        will   1334122
36       their   1269816
37         all   1192059
38        more   1160934
```

```
39         one   1114069
40       about   1109250
41         new   1063545
42       which   1027401
43          if    992077
44          up    979031
45       would    964296
46         can    959020
47        what    931630
48        been    925244
49         had    915768
50        were    915016
51      people    896168
52       there    884623
53          so    875278
54        when    855413
55         out    848854
56         our    829363
57           t    818524
58          no    791847
59         its    757366
60        like    722738
61        your    704146
62        also    692619
63         her    682236
64        than    681516
65          us    676921
66          do    668195
67         she    658402
68       after    656209
69       other    654975
70        time    645617
71        some    642398
72        just    623943
73        over    605479
74         now    601740
75        them    600959
76        into    595788
77          my    594343
78        year    575572
79        only    575520
80          mr    565167
81        even    538672
82         two    537129
83       first    519995
84       state    518669
85         how    514489
86        news    508650
87   president    506203
```

```
88        most    502739
89     because    500331
90       these    494669
91       could    489263
92        many    488385
93       years    481299
94  government    480482
95       trump    479373
96         any    460356
97       obama    458772
98       world    453223
99         may    448478
```

```python
pd.set_option('display.max_rows', 100)
print(top_100_df)
```

```
          word     count
0          the  26197217
1           to  12665105
2           of  12497559
3          and  11515006
4            a  10016572
5           in   8722310
6         that   5601978
7           is   5073471
8          for   4386815
9           on   3522241
10          it   3357914
11           s   3324067
12        with   2834158
13          as   2804255
14         was   2446471
15         are   2444393
16          by   2344952
17           i   2291734
18        this   2281746
19          at   2198249
20          be   2176130
21         not   2114237
22          he   2109271
23         you   2100645
24        have   2099335
25        from   2043114
26          an   1723904
27         has   1709606
28        they   1676544
29         but   1647773
30         his   1632199
31          we   1602266
32          or   1534999
```

33	said	1415584
34	who	1347082
35	will	1334122
36	their	1269816
37	all	1192059
38	more	1160934
39	one	1114069
40	about	1109250
41	new	1063545
42	which	1027401
43	if	992077
44	up	979031
45	would	964296
46	can	959020
47	what	931630
48	been	925244
49	had	915768
50	were	915016
51	people	896168
52	there	884623
53	so	875278
54	when	855413
55	out	848854
56	our	829363
57	t	818524
58	no	791847
59	its	757366
60	like	722738
61	your	704146
62	also	692619
63	her	682236
64	than	681516
65	us	676921
66	do	668195
67	she	658402
68	after	656209
69	other	654975
70	time	645617
71	some	642398
72	just	623943
73	over	605479
74	now	601740
75	them	600959
76	into	595788
77	my	594343
78	year	575572
79	only	575520
80	mr	565167
81	even	538672

```
82          two    537129
83        first    519995
84        state    518669
85          how    514489
86         news    508650
87    president    506203
88         most    502739
89      because    500331
90        these    494669
91        could    489263
92         many    488385
93        years    481299
94   government    480482
95        trump    479373
96          any    460356
97        obama    458772
98        world    453223
99          may    448478
```

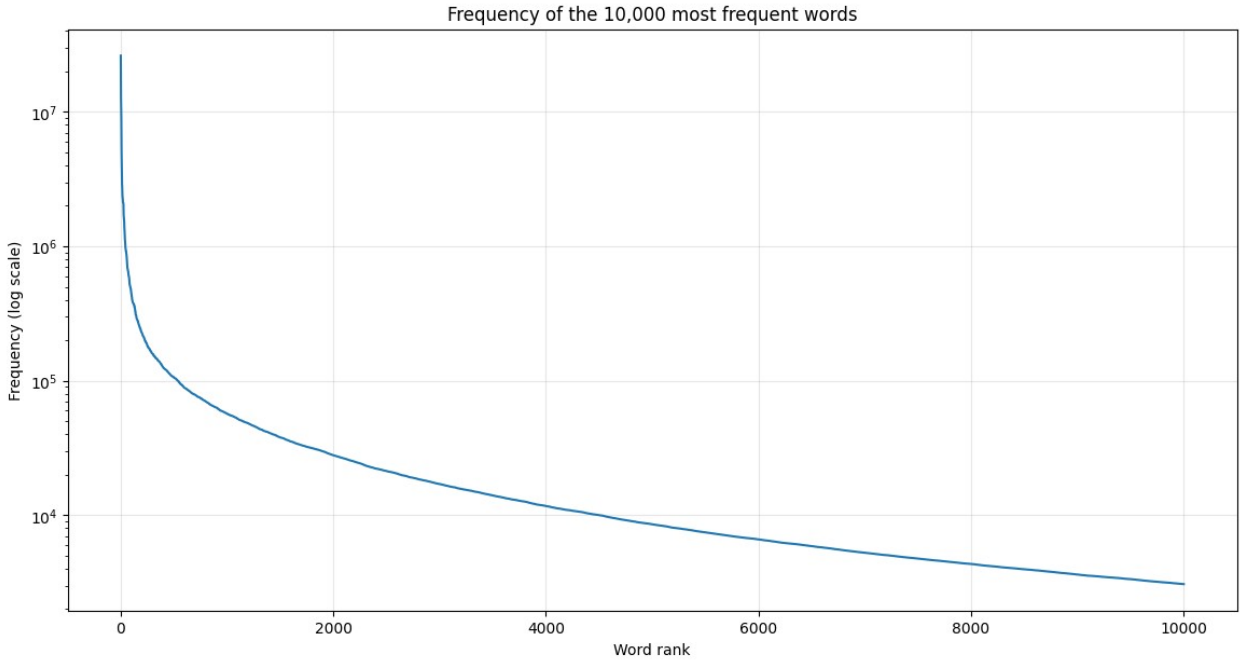
7. Plot the frequency of the 10,000 most frequent words

Because frequencies usually drop very quickly, the y-axis is shown on a logarithmic scale.

```
top_10000 = word_counts.most_common(10000)

freq_values = [count for _, count in top_10000]
ranks = range(1, len(freq_values) + 1)

plt.figure(figsize=(14, 7))
plt.plot(ranks, freq_values)
plt.yscale("log")
plt.xlabel("Word rank")
plt.ylabel("Frequency (log scale)")
plt.title("Frequency of the 10,000 most frequent words")
plt.grid(True, alpha=0.3)
plt.show()
```


Frequency of the 10,000 most frequent words
Frequency (log scale)
10^7
10^6
10^5
10^4
0
2000
4000
6000
8000
10000
Word rank